Numerical verification for harmonic maps

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Abstract

In this working paper, we develop a framework to construct harmonic maps $S^3 \to S^2$ numerically, and use this to construct approximate harmonic maps in homotopy classes with Hopf invariant $2, 3, 4, 5, 6$. By using a rigorous fixed-point argument, we show that there exist genuine harmonic maps with the same Hopf invariant close to our approximate solutions, and thus find a family of counterexamples to Eells' conjecture, and the related Ding's conjecture.
Disclaimer: this paper is a placeholder of the completed version to appear on arXiv.
Code: github.com/edhirst/eells_counterexample

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1 Introduction

Harmonic maps $u : M \to N$ are defined as critical points of the Dirichlet energy functional

$$E(u) = \frac{1}{2} \int_M |du|^2 \, d\mathrm{vol}_M.$$

There are well-known obstacles to finding such critical points: first, often, the infimum of the energy in a fixed topological class is zero (see [41, Section 2.5] for a precise statement). Hence, direct minimisation methods for E cannot produce harmonic maps in these cases. Second, in many interesting situations, harmonic maps are proven to not even be a *local* minimum, but their second variation has a positive index (see e.g. [34, Theorem I.1]).

The result is a large body of research constructing examples of harmonic maps and analysing their obstructions using sophisticated methods: the heat flow method from [18], the min-max constructions in [37, 28, 27], using symmetries and ODE methods in [40, 15, 8], the gluing constructions in [10, 26] and gluing obstructions in [36], and the holomorphic and twistor constructions in [17, 16], to list a few.

On the other hand, a great variety of numerical techniques has been applied to numerically compute harmonic maps. An excellent overview over different approaches is provided in [5]. The main approaches include: finite element methods based on the energy-decreasing iteration of [1], discretised in [4], and the more recent approximation results for stable harmonic maps in [7], or approximating the harmonic-map heat flow in [6]; finite-difference methods, e.g. the recent [49]; lattice discretisations [30] and intrinsic geometric methods such as [21, 22]; the approach using point clouds from [48]; and ODE techniques for symmetry-reduced solutions in [9].

An important special case that has received much attention is the case of harmonic maps between round spheres, and among these the case of $u : S^3 \to S^2$ in particular. In this situation, solutions must be unstable by [34, Theorem I.1]. The Hopf fibration $\pi : S^3 \to S^2$ is a classical example of a harmonic map in this case. More generally, one checks that $R \circ \pi \circ A$ is a harmonic map for $R : S^2 \to S^2$ a weakly conformal map and $A : S^3 \to S^3$ an isometry [2]. To our knowledge, no other harmonics maps $S^3 \to S^2$ have appeared in the literature, neither through with a mathematical proof, nor using numerical experiments. Indeed, Eells's conjecture (see [46, p. 730]) says:

Conjecture 1.1 (Eells)**.** *Every harmonic map $u : S^3 \to S^2$ has the form $u = R \circ \pi \circ A$ for a weakly conformal map $R : S^2 \to S^2$ and an isometry $A : S^3 \to S^3$.*

To a smooth map $u : S^3 \to S^2$ one can assign its (absolute) *Hopf invariant*, see §2.3. The Hopf invariant of the Hopf fibration π is 1, and more generally for a composition $R \circ \pi \circ A$ it is $(\deg A)(\deg R)^2$ [46, Section 2]. Conjecture 1.1 implies the following weaker conjecture, attributed to Ding in [46, p. 730]:

Conjecture 1.2 (Ding)**.** *A homotopy class of maps $S^3 \to S^2$ contains a harmonic map if and only if its (absolute) Hopf invariant is k^2 for some integer k.*

The content of the present paper is the following:

- Rewriting the defining equation for harmonic maps $u : S^3 \to S^2$ as a polynomial equation for maps $u : S^3 \to \mathbb{R}^3$, we can take an ansatz for u as a linear combination of spherical harmonics, and use spectral methods to solve the equation. Apart from the usual good convergence properties of spectral methods, we find that the method is robust and for many initial data converges to approximate harmonic maps with any given Hopf invariant. In particular, we exhibit one numerical example that appears to contradict Conjectures 1.1 and 1.2, though this does not constitute a proof.

- We then apply a fixed-point argument using elliptic PDE techniques to prove that near this numerical example, there exists a genuine harmonic map. We quantify exactly what *near* means, and are able to compute its Hopf invariant and tight enclosures for its energy. This turns the numerical candidate from the previous bullet point into a proven counterexample to the conjectures above.

Our main result is:

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$H(u)$	2	3	4	5	6
$E(u)/8\pi^2$	1.587	1.971	2.251	2.472	2.652

Table 1.1: Hopf invariants and normalised Dirichlet energies $\frac{E}{8\pi^2}$ of our harmonic maps u, rounded to three decimal places.

Theorem 1.1. *There exist harmonic maps $u : S^3 \to S^2$ with the Hopf invariants and energies listed in Table 1.1. None of the maps are of the form $u = R \circ \pi \circ A$ for an isometry $A : S^3 \to S^3$ and a weakly conformal map $R : S^2 \to S^2$.*

In particular, this disproves Conjecture 1.2 and thereby the stronger Conjecture 1.1. It also disproves Conjecture 1.1 for cases with square Hopf invariant.

The technique of constructing an approximate solution to an elliptic PDE and perturbing it to a genuine solution has been a standard tool in geometric analysis since [42] and has since been applied in countless important works. These proofs usually use pen-and-paper techniques. Much more recently, this same approach has been applied in geometry using computer aid in the form of *numerically verified proofs*. In [11, 47, 38, 12], this was carried out using symmetries, reducing the analysis to an ODE. In [33], this was applied to a PDE on the 2-sphere.

In our case, we also impose symmetries on our approximate solution and all function spaces, which guarantee that the genuine solution has the same symmetries. We consider a family of actions of $G = O(2) \times \mathbb{Z}_2$ on S^3 and S^2 (see Section 2.1), searching for equivariant maps. These actions have cohomogeneity two, so we cannot apply the ODE techniques from the references above, but apply general PDE techniques that work in any dimension. However, the reduction to two dimensions provides important speed-ups for our numerical work.

Remark 1.2. The harmonic maps in Conjecture 1.1 have energy $E = 8\pi^2 |\deg R| = 8\pi^2\sqrt{H}$, so the energies listed in Table 1.1 exclude our solutions in theorem 1.1 from being of this form. Moreover, at least when the Hopf invariant is even, one can a-priori exclude any map of the form in Conjecture 1.1 from our equivariant function spaces, see Remark 2.1.

This proof method requires to give precise bounds for all quantities involved: residuals, linear operators, and higher order terms. It requires reducing analysis in infinite-dimensional Banach spaces to finite computations that can be implemented on a computer. While applications in geometry are still few, we build on a solid foundation of existing numerically verified proofs in analysis. We mention the important existence proof of the Lorentz attractor in [45] as an example, and refer to the survey [23] and the textbook [31] for comprehensive overviews.

One important feature of the proof technique is that we obtain very detailed information about the genuine solutions coming from the approximate solutions. Apart from the energies shown in theorem 1.1, we demonstrate this principle with the following result:

Theorem 1.3. *The harmonic map $u : S^3 \to S^2$ with Hopf invariant 2 from theorem 1.1 has* $\operatorname{index}(u) \geq 9$.

There are 4 conformal Killing fields on S^3. Precomposing u with their flows gives negative directions for the second variation of the Dirichlet energy, see [19, Theorem 1.1] and [34, §II.5]. Thus, $\operatorname{index}(u) \geq 4$. Even more: [34, Theorem I.1] proves $\operatorname{index}(u) \geq 5$ if u is not of the form $R \circ \pi \circ A$, and our lower bound strengthens the result in this example.

The paper is structured as follows: in section 2 we make the explicit choices that are used in the later analysis. In particular, in section 2.1 we introduce the spherical harmonics and in section 2.2 we rewrite the harmonic map equation as a *polynomial* equation. Following this, we prove the estimates for the residual, linearised operator, and higher order terms in section 3. The final section section 4 is then a simple matter of assembling the earlier estimates to obtain proofs for theorems 1.1 and 1.3. The appendices list some of the data used in our certified computations.

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2 Background

In this section we marshal some definitions and notations that are used throughout the remainder of the paper. In section 2.1, we define the Sobolev spaces of equivariant maps that we will use for our functional analysis. In section 2.2, we rewrite the harmonic map equation for polynomials. Following this, in section 2.3, we define the Hopf invariant and prove a simple emma that will later allow us to compute the Hopf invariant of our approximately harmonic candidate map using a homotopy. In section 2.4, we introduce the quadratic forms and estimates used to obtain the index lower bound. Last, in section 2.5 we will state a fixed-point theorem that will be applied at the very end to perturb the approximately harmonic map to a genuine harmonic map.

2.1 Sobolev spaces with symmetries

In this subsection, we define Banach spaces of functions that have the symmetries we expect of our solution. We will then carry out all functional analysis in these function spaces and show that a solution indeed exists in them. This approach of defining symmetry-reduced function spaces is standard for elliptic PDEs. An example for harmonic maps is [13]. In our case, we will consider *equivariant* maps $S^3 \to S^2$, hence we begin by defining group actions on S^3 and S^2, or, more precisely, on the ambient $\mathbb{R}^4$ and $\mathbb{R}^3$ that restrict to the spheres.

2.1.1 Definition of the symmetry group.

Identify $\mathbb{R}^4 = \mathbb{C}^2$ and $\mathbb{R}^3 = \mathbb{C} \times \mathbb{R}$. Let $G = O(2) \times \mathbb{Z}_2$, with circle subgroup $e^{i\theta}$, reflection $\kappa \in O(2)$ and generator $\varepsilon \in \mathbb{Z}_2$. For $z, w, \zeta \in \mathbb{C}$ and $t \in \mathbb{R}$, and some positive integer $q > 1$, we define the action of G on $\mathbb{R}^4$ and $\mathbb{R}^3$ by the following,

$$\begin{aligned} e^{i\theta} \cdot (z, w) &= (e^{i\theta} z, e^{-qi\theta} w), & e^{i\theta} \cdot (\zeta, t) &= (e^{-2qi\theta} \zeta, t), \\ \kappa \cdot (z, w) &= (\bar{z}, \bar{w}), & \kappa \cdot (\zeta, t) &= (\bar{\zeta}, t), \\ \varepsilon \cdot (z, w) &= (z, -w), & \varepsilon \cdot (\zeta, t) &= (\zeta, -t). \end{aligned} \tag{2.1}$$

All these orthogonal actions restrict to the round unit spheres S^3 and S^2. We also define the following spaces of equivariant functions:

$$\begin{aligned} X &:= H^2_G(S^3, \mathbb{R}^3) := \{u \in H^2(S^3, \mathbb{R}^3) : u(gx) = gu(x) \text{ for all } g, x\}, \\ Y &:= L^2_G(S^3, \mathbb{R}^3) := \{u \in L^2(S^3, \mathbb{R}^3) : u \circ g = gu \text{ in } L^2 \text{ for all } g \in G\}, \end{aligned}$$

and endow them with the norms $\|u\|_X = \|(1 - \Delta)u\|_{L^2}$ and $\|u\|_Y = \|u\|_{L^2}$. Here, and throughout the text, we will refer to the group G as G_q when we want to specify that it acts via (2.1) for some given q. Otherwise, we will just refer to X, Y and G when our analysis is independent of q.

Remark 2.1. When q is even, the action of $(e^{i\pi}, \varepsilon) \in G_q$ in (2.1) is given by

$$(z, w) \mapsto (-z, -w) \qquad\qquad (\zeta, t) \mapsto (\zeta, -t)$$

This excludes a G_q-equivariant map $u : S^3 \to S^2$ being of the form $R \circ \pi \circ A$, where $R : S^2 \mapsto S^2$ is surjective, $\pi : S^3 \to S^2$ is the Hopf projection, and $A \in O(4)$. One can argue as follows: since $\pi(-z, -w) = \pi(z, w)$ and $A(-z, -w) = -A(z, w)$ by linearity, then $u(z, w) = R \circ \pi \circ A(z, w) = u(-z, -w)$. If u also equivariant, then the image of u must lie in the plane $\mathbb{C}$ fixed by $(\zeta, t) \mapsto (\zeta, -t)$, so R cannot be surjective. In particular, for q even, this rules out harmonic maps of this form occurring in our function spaces X and Y. Indeed, our previous argument shows that u would be a harmonic map $S^3 \to S^1$, which must be constant.

2.1.2 Spherical harmonics.

Throughout the non-rigorous initial numerical work as well as for the later functional analysis, we crucially use the spherical harmonics on S^3. For optimisation, they allow us to use the spectral Galerkin methods

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described in section 3.1. For functional analysis, they allow us to decompose the linearised elliptic operator into blocks that can be computed effectively. We use the non-positive Laplacian $\Delta = \operatorname{div}\nabla$. The complex spherical harmonics of degree l are the restrictions to S^3 of complex homogeneous harmonic polynomials of degree l on $\mathbb{R}^4$. They form the $(l+1)^2$-dimensional eigenspace of $-\Delta$ with eigenvalue $l(l+2)$. We use the Jacobi basis of [29, §2, (4)], normalized for volume $2\pi^2$:

$$Y_{m,n,j}(z,w) = c_{|m|,|n|,j}\, z^{(m)} w^{(n)} P_j^{(|m|,|n|)}(1-2|z|^2), \tag{2.2}$$

$$c^2_{a,b,j} = \frac{(2j+a+b+1)(j+a+b)!\, j!}{2\pi^2 (j+a)!(j+b)!}, \qquad l = |m|+|n|+2j. \tag{2.3}$$

Here $m, n \in \mathbb{Z}$, $j \geq 0$, and $z^{(m)} = z^m$ for $m \geq 0$, $z^{(m)} = \bar{z}^{|m|}$ for $m < 0$, similarly for $w^{(n)}$. Moreover $P_j^{(a,b)}$ is the Jacobi polynomial with the real variable $(1-2|z|^2)$, i.e.

$$P_j^{(a,b)}(1-2|z|^2) = \sum_{s=0}^{j} (-1)^s \binom{j+a}{j-s} \binom{j+a+b+s}{s} |z|^{2s}.$$

For each degree l, the functions $Y_{m,n,j}$ with (m,n,j) such that $|m|+|n|+2j = l$ form an L^2-orthonormal basis of the Laplacian eigenspace.

Lemma 2.2. *The action of G_q on $Y_{m,n,j}(z,w)$ is as follows:*

1. $e^{i\theta} \cdot Y_{m,n,j} = e^{i(m-qn)\theta} Y_{m,n,j}$
2. $\kappa \cdot Y_{m,n,j} = Y_{-m,-n,j} = \overline{Y_{m,n,j}}$
3. $\varepsilon \cdot Y_{m,n,j} = (-1)^n Y_{m,n,j}$

We now specialise to the *equivariant* subset of our basis, where the target $\mathbb{R}^3$ is now also equipped with a G-action (2.1), and also describe *G-invariant* subset.

Proposition 2.3. *Let $G = G_q$. The following fields are L^2-orthonormal and their linear spans are dense in X and Y with respect to their respective norms:*

$$(\operatorname{Re} Y_{q(n-2),n,j}, \operatorname{Im} Y_{q(n-2),n,j}, 0), \qquad n \text{ even}, \quad j \geq 0,$$
$$(0, 0, \sqrt{2} \operatorname{Re} Y_{qn,n,j}), \qquad n \geq 1 \text{ odd}, \quad j \geq 0.$$

Moreover, the real scalar functions

$$Y_{0,0,j} \quad j \geq 0, \qquad \sqrt{2}\operatorname{Re} Y_{qn,n,j} \quad n \geq 2 \text{ even}, \ j \geq 0.$$

are orthonormal and their linear span is dense in the space of G_q-invariant L^2-functions on S^3.

Proof. We identify $\mathbb{R}^3 = \mathbb{C} \times \mathbb{R}$ as before. By definition, this splitting is equivariant with respect to the G action. We first determine which members of the Jacobi basis $Y_{m,n,j}$ are equivariant for each component. In the first component $\mathbb{C}$, part (1) of Lemma 2.2 gives $m - qn = -2q$, and part (3) gives that n must be even. The second component $\mathbb{R}$ is similar, only $\varepsilon \in G$ acts non-trivially on the target as -1, so we must have $m = qn$, and n odd. The restriction $n \geq 1$ avoids the overcounting coming from (2), since $\operatorname{Re} Y_{qn,n,j} = \operatorname{Re} Y_{-qn,-n,j}$.

The statement for the G-invariant subset is similar, but now with a trivial action on the target. The same argument as before gives $m = qn$, and n even. Note that $Y_{0,0,j}$ is automatically real by Lemma 2.2. □

Enumerate the vector basis by (Ψ_k) and denote its degrees by l_k. Then for some (m_k, n_k, j) we have $l_k = |m_k| + |n_k| + 2j_k$, which gives $l_k \geq 2$ by Proposition 2.3. Moreover, we have that

$$\Big\| \sum_k a_k \Psi_k \Big\|_X^2 = \sum_k (l_k+1)^4 |a_k|^2.$$

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2.1.3 Sobolev norms.

In addition to the X-norm, we use the Sobolev norm $\|\xi\|_{H^1}^2 = \|\xi\|_{L^2}^2 + \|\nabla\xi\|_{L^2}^2$. We have the following Sobolev embedding theorems for the chosen Sobolev norms:

Proposition 2.4. *For $u \in X$ and $\xi \in H^1(S^3, \mathbb{R}^3)$,*

$$\|u\|_{C^0} \le c_\infty \|u\|_X, \quad \|\nabla u\|_{L^4} \le c_4 \|u\|_X, \quad \|\Delta u\|_{L^2} \le \|u\|_X, \quad \|\xi\|_{L^4} \le \beta \|\xi\|_{H^1}, \quad \|\xi\|_{L^4}\|\nabla\xi\|_{L^2} \le \beta \|\xi\|_{H^1}^2$$

where

$$c_\infty = \sqrt{\frac{\pi^2/6 - 5/4}{2\pi^2}}, \qquad c_4 = \left(\frac{2}{9\pi^2}\right)^{1/4}, \qquad \beta = (2\pi^2)^{-1/4}.$$

Proof. It suffices to consider finite harmonic sums $u = \sum_{l\ge 2} u_l$, where u_l has degree l. The addition formula [14, Corollary 1.2.7], applied componentwise, and Cauchy–Schwarz give

$$\|u\|_{C^0} \le \frac{1}{\sqrt{2\pi^2}} \sum_{l\ge 2} (l+1)\|u_l\|_{L^2} \le \left(\frac{1}{2\pi^2} \sum_{l\ge 2} \frac{1}{(l+1)^2}\right)^{1/2} \|u\|_X = c_\infty \|u\|_X.$$

The Laplacian estimate follows from $l(l+2) \le (l+1)^2$.

The sharp interpolation inequality on S^3 [3, Eqn. 6.8.7], after rescaling the measure, gives

$$\|f\|_{L^4}^2 \le \frac{1}{\sqrt{2\pi^2}} \left(\|f\|_{L^2}^2 + \frac{2}{3}\|\nabla f\|_{L^2}^2\right) \qquad (f \in H^1(S^3)),$$

which immediately gives the two claims for ξ. Applying this to $f = |\nabla u|$, Kato's inequality and the integrated Bochner formula on S^3 give

$$\|\nabla u\|_{L^4}^2 \le \frac{\|\nabla u\|_{L^2}^2 + \frac{2}{3}\|\nabla^2 u\|_{L^2}^2}{\sqrt{2\pi^2}} = \frac{2\|\Delta u\|_{L^2}^2 - \|\nabla u\|_{L^2}^2}{3\sqrt{2\pi^2}} \le c_4^2 \|u\|_X^2.$$

The estimates then follow from density of finite harmonic sums. □

The space X consists of H^2-functions, so its norm controls the L^2-norm of first derivatives. We record the following explicit bound for this:

Proposition 2.5. *For $u \in X$,*

$$\|du\|_{L^2} \le c_5 \|u\|_X$$

where

$$c_5 = \tfrac{2\sqrt{2}}{9}.$$

Proof. Let $u = \sum_k a_k \Psi_k$. Then

$$\|du\|_{L^2}^2 = \sum_k \frac{l_k(l_k+2)}{(l_k+1)^4} (l_k+1)^4 a_k^2 \le \Big(\max_{l\ge 2} \frac{l(l+2)}{(l+1)^4}\Big) \sum_k (l_k+1)^4 a_k^2 = \frac{8}{81}\|u\|_X^2.$$

□

2.2 The harmonic map equation

A smooth map $u : S^3 \to S^2$ is harmonic if it is a critical point of the Dirichlet energy

$$E(u) = \frac{1}{2}\int_{S^3} |du|^2 \, d\mathrm{vol}.$$

This is the case if and only if it satisfies the elliptic equation

$$\tau(u) = \Delta u + |du|^2 u = 0, \tag{2.4}$$

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see, with the opposite Laplacian convention, [20, (1.12)–(1.13)].

Throughout the paper, we will solve the following equation, which is particularly amenable for numerically verified proofs. For fixed $\bar{W} \in X$ with $\bar{W}$ nowhere zero define

$$\begin{aligned} F : X &\longrightarrow Y, \\ v &\longmapsto \frac{1}{2}\bar{W} \times \Big(W \times \big(3|W|^2\Delta W - \Delta(|W|^2 W)\big)\Big) + |\bar{W}|^2\big(\bar{W}\cdot(1-\Delta)v\big)\bar{W}, \end{aligned} \tag{2.5}$$

where $W = \bar{W} + v$. One checks that F indeed maps equivariant functions to equivariant functions and, using the Sobolev embedding theorems, that it maps H^2-fields to L^2-fields.

Proposition 2.6. *If $v \in X$ satisfies $\|v\|_{C^0} < \min_{S^3} |\bar{W}|$ and $F(v) = 0$, then*

$$u = \frac{\bar{W} + v}{|\bar{W} + v|}$$

is a smooth G-equivariant harmonic map.

Proof. Our assumption $\|v\|_{C^0} < \min_{S^3} |\bar{W}|$ ensures $W = \bar{W} + v$ is non-vanishing, so u is well-defined. We have

$$u \times du = \frac{W \times dW}{|W|^2}, \qquad \operatorname{div}(u \times du) = u \times \Delta u, \qquad \operatorname{div}(W \times dW) = W \times \Delta W. \tag{2.6}$$

This gives

$$\begin{aligned} |W|^4 u \times \Delta u &= |W|^4 \operatorname{div}\left(\frac{W \times dW}{|W|^2}\right) \\ &= W \times \big(|W|^2\Delta W - \langle d(|W|^2), dW\rangle\big) \\ &= \frac{1}{2} W \times \big(3|W|^2\Delta W - \Delta(|W|^2 W)\big), \end{aligned}$$

where in the first step we used (2.6); in the second step we used the product rule, $d(|W|^{-2}) = -|W|^{-4} d(|W|^2)$, and (2.6); and in the third step we used the product rule for $\Delta(|W|^2 W)$ and $W \times W = 0$.

Plugging this into the definition of F gives

$$0 = F(v) = |W|^4 \bar{W} \times (u \times \Delta u) + |\bar{W}|^2\big(\bar{W}\cdot(1-\Delta)v\big)\bar{W}.$$

The two summands are orthogonal, so $F(v) = 0$ implies $u \times \Delta u = 0$. Here we used that $\bar{W} \cdot u > 0$ and $u \times \Delta u$ is perpendicular to u. Differentiating $|u|^2 = 1$ gives

$$0 = \frac{1}{2}\Delta(|u|^2)u = u \times (u \times \Delta u) + \Delta u + |du|^2 u = \Delta u + |du|^2 u,$$

which is (2.4). Since $du \in L^6$, standard elliptic theory gives that u is smooth. □

Remark 2.7. As the proof of Proposition 2.6 shows, the two terms in (2.5) vanish independently. The first term comes from the harmonic map equation (2.4), re-written for $u = W|W|^{-1}$, while the second term is a gauge-fixing, which deals with the freedom to rescale W by a function to give the same harmonic map u. In this case, the second term in (2.5) fixes the component of $W = \bar{W} + v$ along $\bar{W}$ while the first term fixes the component of W orthogonal to $\bar{W}$.

2.3 The Hopf invariant

Definition 2.8. Let ω be an area form on S^2 normalized by $\int_{S^2} \omega = 1$. Since $H^2_{dR}(S^3) = 0$, for any smooth map $f : S^3 \to S^2$, one can choose $\alpha \in \Omega^1(S^3)$ with $d\alpha = f^*\omega$. We define the (absolute) *Hopf invariant* by

$$H(f) = \Big|\int_{S^3} \alpha \wedge d\alpha\Big| \in \mathbb{Z}_{\geq 0}.$$

This is independent of α, invariant under homotopy, and extends to continuous maps by smooth approximation.

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One can compute the Hopf invariant in many cases of interest using the following proposition, proved in [25, Theorem 2]:

Proposition 2.9. *Suppose $F : S^3 \to S^3$, $f : S^3 \to S^2$ and $G : S^2 \to S^2$ are smooth, then*

$$H(G \circ f \circ F) = (\deg G)^2 \, |\deg F| \, H(f).$$

This is particularly useful combined with the following map:

Definition 2.10. Identify $\mathbb{R}^4 = \mathbb{C}^2$ and $\mathbb{R}^3 = \mathbb{C} \times \mathbb{R}$. For $z, w \in \mathbb{C}$, define the (extended) *Hopf fibration* $\pi : \mathbb{R}^4 \to \mathbb{R}^3$ as the map

$$\pi(z, w) = (w^2 - \bar{z}^2, 2\operatorname{Re}(zw)) \tag{2.7}$$

This restricts to a map $\pi : S^3 \to S^2$ on the sphere $|z|^2 + |w|^2 = 1$.

As a map $\pi : S^3 \to S^2$, [25, Theorem 2] shows that the Hopf fibration satisfies $H(\pi) = 1$. This has the following consequence

Lemma 2.11. *Suppose $F : S^3 \to \mathbb{R}^4$ with $|F|$ non-zero. Let $V = \pi(F)$, where π is the extended Hopf fibration. Then $\frac{V}{|V|}$ is a map $S^3 \to S^2$, with*

$$H\left(\tfrac{V}{|V|}\right) = \deg\left(\tfrac{F}{|F|}\right).$$

Proof. This uses two key properties of the extended Hopf map π, which are easy to verify. These are that $|\pi(z, w)| = |z|^2 + |w|^2$, and that $\pi(\lambda z, \lambda w) = \lambda^2 \pi(z, w)$ for any $\lambda > 0$. In particular

$$\frac{V}{|V|} = \frac{V}{|F|^2} = \pi\left(\frac{F}{|F|}\right)$$

We now apply Proposition 2.9 and conclude. □

Later on, we will show that our approximate harmonic map is homotopic to a reference map with a known Hopf invariant, which will allow us to compute the Hopf invariant of the approximate solution.

The reference map we will use is the following G_q-equivariant map:

$$V_q(z, w) := \left(w^2 - \tfrac{1}{4}\bar{z}^{2q}, \operatorname{Re}(z^q w)\right) \tag{2.8}$$

Some properties of this reference map follow by basic calculation

Lemma 2.12. *The map V_q in (2.8) satisfies $|V_q| = \frac{1}{4}|z|^{2q} + |w|^2$. In particular, restricted to sphere $|z|^2 + |w|^2 = 1$, we have the inequality $0 < |V_q| \leq 1$.*

Proof. The lower bound is trivial. On the sphere $|z|^2 \leq |z|^2 + |w|^2 = 1$ so $\frac{1}{4}|z|^{2q} + |w|^2 \leq |z|^2 + |w|^2 = 1$, which proves the upper bound. □

Since V_q is non-vanishing, we may consider $\frac{V_q}{|V_q|}$ as a map $S^3 \to S^2$, and use Lemma 2.11 to compute its Hopf invariant.

Proposition 2.13. $H\left(\frac{V_q}{|V_q|}\right) = q$

Proof. Up to scaling z by a constant (which does not change the homotopy class), V_q is precisely $\pi(z^q, w)$. The map

$$(z, w) \mapsto (z^q, w)$$

suitably normalised, is a q-fold (branched) covering map on S^3, since on the regular locus $z \neq 0$, the pre-image of a point (z, w) is generated by the qth root of unity. Hence, the degree of this map is precisely q. The proposition now follows by Lemma 2.11. □

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Remark 2.14. The z-scaling factor in V_q was chosen arbitrarily in order to ensure upper bound of Lemma 2.12 holds, while the ensuring the numerical homotopy lower bound in Proposition 2.15 below was sufficiently large, see also Proposition 3.1.

The following proposition reduces the construction of a homotopy to exhibiting a suitable vector field:

Proposition 2.15. *Suppose that $W_0, W_1 : S^3 \to \mathbb{R}^3$ are continuous and*

$$W_0 \cdot W_1 > 0 \qquad \text{on } S^3.$$

Then W_0, W_1 are nowhere zero, and $\frac{W_0}{|W_0|}$ is homotopic to $\frac{W_1}{|W_1|}$. In particular, $H(\frac{W_0}{|W_0|}) = H(\frac{W_1}{|W_1|})$.

Proof. Firstly, $|W_0|, |W_1|$ must be non-vanishing by Cauchy-Schwarz, as is $|(1-t)W_0 + tW_1|$ for $t \in [0,1]$. Thus, the following defines a homotopy between $W_0/|W_0|$ and $W_1/|W_1|$ as maps $S^3 \to S^2$:

$$W_t = \tfrac{(1-t)W_0 + tW_1}{|(1-t)W_0 + tW_1|}$$

□

2.4 Index of a harmonic map

The second variation of the Dirichlet energy functional, at a critical point $u : S^3 \to S^2$ with $\tau(u) = 0$ is given by the quadratic form:

$$Q_u(\xi) = \int_{S^3} |d\xi|^2 - |du|^2 |\xi|^2 \tag{2.9}$$

where $\xi \cdot u = 0$ and $d\xi$ is the component-wise derivative of ξ in $\mathbb{R}^3$. This quadratic form is defined on sections $\Gamma_{H^1}(u^*TS^2) = \{\xi \in H^1(S^3, \mathbb{R}^3) : \xi \cdot u = 0\}$.

In this section, we will work with the following quadratic form, which is better adapted to our set up. For fixed $W \in X$ with W nowhere zero, and $\xi \in H^1(S^3, \mathbb{R}^3)$, define $Q_W : H^1(S^3, \mathbb{R}^3) \to \mathbb{R}$ as

$$Q_W(\xi) := \int_{S^3} \left| d\xi - \frac{(W \times dW) \times \xi}{|W|^2} \right|^2 - \frac{|(W \times dW) \cdot \xi|^2}{|W|^4} + \frac{(W \cdot \xi)^2}{|W|^2}. \tag{2.10}$$

Proposition 2.16. *Let Q_u, Q_W, be the operators in (2.9), (2.10) acting on $\Gamma_{H^1}(u^*TS^2) = \{\xi \in H^1(S^3, \mathbb{R}^3) : \xi \cdot u = 0\}$ and $H^1(S^3, \mathbb{R}^3)$ respectively, and suppose $u = \frac{W}{|W|}$. Then:*

$$\mathrm{Ind} Q_u = \mathrm{Ind} Q_W \qquad\qquad \ker Q_u = \ker Q_W$$

Proof. Let $P_u = I - uu^T$ be the projection to the orthogonal complement of u. We define Q_W in (2.10) such that

$$Q_W(\xi) = Q_u(P_u\xi) + \int_{S^3} |d(u \cdot \xi)|^2 + (u \cdot \xi)^2 . \tag{2.11}$$

From this, one can see that Q_u and Q_W agree on $P_u(\xi)$, the component of ξ orthogonal to u. They differ only the component parallel to u, but by a positive-definite form. The parallel summand is positive definite, so it contributes no index or kernel. □

The point of doing this is so that we can pass to a quadratic form over arbitrary maps ξ into $\mathbb{R}^3$, just like we did when we gauge-fixed the components in the $\bar{W}$-directions in equation (2.5), see Remark 2.7.

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2.4.1 Estimating the index

Later on we will use an approximate harmonic map $\bar{W}$ and perturb it to a genuine harmonic map W. To ensure that negative directions for $Q_{\bar{W}}$ remain negative for Q_W, we estimate the difference between these quadratic forms.

Proposition 2.17. *Let $W, \bar{W} \in X$, $r, C_1 > 0$ with $\|\bar{W} - W\|_X \le r$ and $\min_{S^3} |\bar{W}| - c_\infty r \ge C_1 > 0$. Then, for $\xi \in H^1(S^3, \mathbb{R}^3)$, $|Q_W(\xi) - Q_{\bar{W}}(\xi)| \le \delta \|\xi\|_{H^1}^2$ with*

$$\delta = r\Big[2\beta\Big(\frac{3c_\infty c_4(\|\bar{W}\|_X + r)}{C_1^2} + \frac{c_4}{C_1}\Big) + \frac{12\beta^2 c_\infty c_4^2(\|\bar{W}\|_X + r)^2}{C_1^3} + \frac{4\beta^2 c_4^2(\|\bar{W}\|_X + r)}{C_1^2} + \frac{4c_\infty}{C_1}\Big].$$

where β, c_4, c_∞ are the constants in Proposition 2.4.

Proof. Write $W = \bar{W} + v$ with $\|v\|_X \le r$, and define

$$A := \frac{W \times dW}{|W|^2}, \qquad u := \frac{W}{|W|},$$

with $\bar{A}$, $\bar{u}$ defined similarly for $\bar{W}$. We need to bound

$$\begin{aligned} Q_W(\xi) - Q_{\bar{W}}(\xi) = \int -2\langle d\xi, (A - \bar{A}) \times \xi\rangle + \big(|A \times \xi|^2 - |\bar{A} \times \xi|^2\big) \\ - \big(|A \cdot \xi|^2 - |\bar{A} \cdot \xi|^2\big) + \big((u \cdot \xi)^2 - (\bar{u} \cdot \xi)^2\big). \end{aligned} \tag{2.12}$$

in terms of $\|\xi\|_{H^1}$.

We can factorise the last three terms in (2.12) as the *difference of squares*: e.g. for the third term in the integrand of (2.12),

$$|A \cdot \xi|^2 - |\bar{A} \cdot \xi|^2 = \big((A - \bar{A}) \cdot \xi\big) \cdot \big((A + \bar{A}) \cdot \xi\big).$$

Using this factorisation, we see that the integrand in (2.12) is bounded in absolute value by

$$2|A - \bar{A}|\,|\xi||d\xi| + 2|A - \bar{A}|\big(|A| + |\bar{A}|\big)|\xi|^2 + |u + \bar{u}||u - \bar{u}|\,|\xi|^2. \tag{2.13}$$

In the following, we define the following functions on S^3:

$$T_1 := 2|A - \bar{A}| \qquad T_2 := 2|A - \bar{A}|\big(|A| + |\bar{A}|\big) + |u + \bar{u}||u - \bar{u}| \tag{2.14}$$

Our proof strategy is as follows: we will apply Hölder's inequality with three exponents $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$ to the integral of (2.13):

$$\int_{S^3} T_1|\xi||d\xi| + T_2|\xi|^2 \le \|T_1\|_{L^4}\|\xi\|_{L^4}\|d\xi\|_{L^2} + \|\xi\|_{L^4}^2\|T_2\|_{L^2} \le \big(\beta\|T_1\|_{L^4} + \beta^2\|T_2\|_{L^2}\big)\|\xi\|_{H^1}^2$$

where we have used Proposition 2.4 in the second inequality. Since the integral of (2.13) bounds (2.12), this gives us the bound $|Q_W(\xi) - Q_{\bar{W}}(\xi)| \le \delta\|\xi\|_{H^1}^2$ in the statement of the proposition if we can bound $\|T_1\|_{L^4}$, and $\|T_2\|_{L^2}$.

To bound $\|T_1\|_{L^4}$, and $\|T_2\|_{L^2}$, we first give a pointwise bound on T_1, T_2, which we will combine with the global bound $|v| \le c_\infty r$ (coming from Proposition 2.4) where appropriate.

1. Bounds on u. Since $|u| = |\bar{u}| = 1$ by definition, $|u + \bar{u}| \le 2$. Using $|a/|a| - b/|b|| \le 2|a - b|/|a|$ for nonzero a, b, we have that
$$|u - \bar{u}| \le \frac{2|v|}{C_1} \le \frac{2c_\infty r}{C_1}$$

2. Bounds on A. Since $|W \times dW| \le |W||dW|$, we have that
$$|A| \le \frac{|dW|}{C_1} \qquad |A| + |\bar{A}| \le 2\frac{|d\bar{W}| + |dv|}{C_1}.$$

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For $A - \bar{A}$, let $W_t := \bar{W} + tv$ and differentiate $A_t = W_t \times dW_t/|W_t|^2$ in t:

$$\dot{A}_t = \frac{v \times dW_t + W_t \times dv}{|W_t|^2} - \frac{2(W_t \cdot v)\, W_t \times dW_t}{|W_t|^4},$$

Since $|dW_t| \le |d\bar{W}| + |dv|$, so $|\dot{A}_t| \le \dfrac{3|v|\left(|d\bar{W}| + |dv|\right)}{C_1^2} + \dfrac{|dv|}{C_1}$. Integrating over $t \in [0,1]$ gives

$$|A - \bar{A}| \le \frac{3|v|\left(|d\bar{W}| + |dv|\right)}{C_1^2} + \frac{|dv|}{C_1} \le \frac{3rc_\infty\left(|d\bar{W}| + |dv|\right)}{C_1^2} + \frac{|dv|}{C_1}$$

Equipped with these pointwise estimates, we can finally bound $\|T_1\|_{L^4}$, since, by Proposition 2.4:

$$\|dv\|_{L^4} \le c_4 r \qquad\qquad \||d\bar{W}| + |dv|\|_{L^4} \le c_4(\|\bar{W}\|_X + r) \tag{2.15}$$

Similarly for $\|T_2\|_{L^2}$, we use Proposition 2.4 to conclude:

$$\||dv|(|d\bar{W}| + |dv|)\|_{L^2} \le \|dv\|_{L^4}\||d\bar{W}| + |dv|\|_{L^4} \le rc_4^2(\|\bar{W}\|_X + r) \tag{2.16}$$

where in the second step, we use Hölders inequality with exponents $\frac{1}{2} = \frac{1}{4} + \frac{1}{4}$. Combining these estimates shows that we can indeed take δ as stated. □

We leverage Proposition 2.17 in the following way. Let $Q(\xi, \eta)$ be the symmetric bilinear form built out of the quadratic form Q, i.e.

$$Q(\xi, \eta) := \tfrac{1}{2}\left[Q(\xi + \eta) - Q(\xi) - Q(\eta)\right].$$

Let $\xi_1, \dots, \xi_k \in H^1(S^3, \mathbb{R}^3)$, and let G, H be the $k \times k$ symmetric matrices with entries:

$$G_{ab} = Q_{\bar{W}}(\xi_a, \xi_b), \qquad H_{ab} = \int_{S^3} \xi_a \cdot \xi_b + \langle d\xi_a, d\xi_b \rangle.$$

Lemma 2.18. *Under the assumptions of proposition 2.17, let $\xi_1, \dots, \xi_k \in H^1(S^3, \mathbb{R}^3)$ and let G, H be as above. If $G + \delta H$ is negative definite, then Q_W is negative definite on their span. In particular* $\operatorname{Ind} Q_W \ge k$.

Proof. Let $c \in \mathbb{R}^k \setminus \{0\}$ and $\xi = \sum_a c_a \xi_a$. By bilinearity, $Q_{\bar{W}}(\xi) = c^T G c$ and $\|\xi\|_{H^1}^2 = c^T H c$, so proposition 2.17 gives

$$Q_W(\xi) \le Q_{\bar{W}}(\xi) + \delta\|\xi\|_{H^1}^2 = c^T(G + \delta H)c < 0.$$

In particular, the ξ_a are linearly independent. □

2.5 The fixed point theorem

Our existence proof proceeds by constructing an explicit approximate solution, and then proving by means of a fixed-point argument that a nearby exact solution exists. This standard approach commonly uses the Newton-Kantorovich theorem as a sufficient criterion for the existence of the true solution, see e.g. [39, Theorem 1]. For simpler statements, we use the following straightforward consequence of the Banach fixed point theorem:

Theorem 2.19. *Let $F : X \to Y$ be a continuously differentiable map between Banach spaces, and suppose $\mathscr{L} = DF(0)$ is invertible. Set $\mathscr{N}(v) = F(v) - F(0) - \mathscr{L}v$. Suppose that, for some $r > 0$,*

$$\|\mathscr{L}^{-1}\|_{Y \to X} \le M, \qquad \|F(0)\|_Y \le \varepsilon, \qquad \|\mathscr{N}(v) - \mathscr{N}(w)\|_Y \le k\|v - w\|_X$$

whenever $\|v\|_X, \|w\|_X \le r$. If $Mk < 1$ and $M(\varepsilon + kr) \le r$, then F has a unique zero in the closed ball of radius r about 0.

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3 Estimates

3.1 Computing the approximate solution

Using the basis from proposition 2.3, we aim to find $\bar{W} = \sum_{l_k \le D} a_k \Psi_k$ solving

$$\left\langle \Psi_k, F(0) + 10(|\bar{W}|^2 - 1)\bar{W} \right\rangle_{L^2} = 0, \qquad l_k \le D.$$

Here, the two terms are pointwise orthogonal, since $\bar{W} \cdot F(0) = 0$, and the second penalises deviations from unit length.

The integrands in this system and its Jacobian are G_q-invariant polynomials of degree at most $6D + 2$. We evaluate these integrals using the exact quadrature from proposition D.1. We then approximately solve the system using damped Newton iteration. Quadrature evaluation and Newton iteration use non-rigorous double precision arithmetic. We point out that there is no theoretical guarantee that the iteration converges to a (truncated) true solution, or even that the non-rigorous arithmetic gives results anywhere near the rigorous values. All rigour comes from the following a posteriori analysis.

The degrees and leading coefficients are given in tables A.1 and B.1. The stored floating-point coefficients are treated as exact rational numbers in the subsequent verification.

3.2 Bounds for the approximate solution

As our approximate solution, take the polynomial field $\bar{W} : S^3 \to \mathbb{R}^3$ as given in the appendix, for a given q. Let $V = V_q$ be the homotopy reference map (2.8) in §2.3.

Proposition 3.1. *For r as in table A.1, the approximate solution $\bar{W}$ satisfies*

$$\begin{aligned}
\|F(0)\|_Y &\le C_2, \\
C_3 \le E\left(\frac{\bar{W}}{|\bar{W}|}\right) &\le C_4, \\
\min_{S^3} \bar{W} \cdot V_q &\ge C_5 > 0, \\
\|\bar{W}\|_X &\le C_6, \\
\|\bar{W}\|_{C^0} &\le C_7, \\
\left\| |\bar{W}|^4 - 1 \right\|_{C^0} &\le \eta < 1, \\
\min_{S^3} |\bar{W}| - c_\infty r &\ge C_1 > 0,
\end{aligned}$$

where the constants are given in table A.1.

Proof. Integrals of G_q-invariant polynomials are computed using proposition D.1. Suprema and infima are bounded using Lipschitz method (see e.g. [32, p. 312, equation (4)] for a description of this standard approach): evaluate on a finite grid and control values between grid points using derivative bounds computed using a Chebyshev–Fourier expansion. All computations are carried out using interval arithmetic. This suffices to compute C_2, C_5, C_7, η, C_1.

The constant C_6 is computed using $\|\bar{W}\|_X^2 = \sum_{l_k \le D} (l_k + 1)^4 a_k^2$.

For C_3, C_4, use $N = |\bar{W} \times d\bar{W}|^2$ and

$$\left| E\left(\frac{\bar{W}}{|\bar{W}|}\right) - \frac{1}{2}\int_{S^3} N(2 - |\bar{W}|^4) \right| = \frac{1}{2}\int_{S^3} \frac{N(|\bar{W}|^4 - 1)^2}{|\bar{W}|^4} \le \frac{\eta^2}{2(1-\eta)} \int_{S^3} N,$$

where the final integral is again computed using proposition D.1. □

Proposition 3.2. *There exist nine linearly independent fields $\xi_1, \ldots, \xi_9 \in H^1(S^3, \mathbb{R}^3)$, orthogonal for both $Q_{\bar{W}}$ and $\langle \cdot, \cdot \rangle_{H^1}$, such that*

$$\max_{1 \le a \le 9} \frac{Q_{\bar{W}}(\xi_a)}{\|\xi_a\|_{H^1}^2} \le \lambda^* < 0,$$

where λ^* is given in table A.1.

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Proof. Let $f_1 \in S_{0,-}$, $f_2 \in S_{1,-}$, $f_3 \in S_{2,-}$, $f_4 \in S_{2,+}$ and $f_5 \in S_{3,-}$ be the nonzero elements of $H^1(S^3, \mathbb{C}^3)$ given in table C.1. These are finite combinations of the P, M, T, with f_1 real. Take the nine real maps

$$\xi_1 = f_1, \qquad \xi_{2a-2} = \operatorname{Re} f_a, \quad \xi_{2a-1} = \operatorname{Im} f_a \qquad (a = 2, \dots, 5).$$

By section C, these are nonzero and pairwise orthogonal for both $Q_{\bar{W}}$ and $\langle \cdot, \cdot \rangle_{H^1}$, hence linearly independent.

By abuse of notation, we denote the extension of both quadratic forms to complex fields by the same symbols. For $a = 2, \dots, 5$, section C gives

$$\frac{Q_{\bar{W}}(f_a)}{\|f_a\|_{H^1}^2} = \frac{Q_{\bar{W}}(\operatorname{Re} f_a)}{\|\operatorname{Re} f_a\|_{H^1}^2} = \frac{Q_{\bar{W}}(\operatorname{Im} f_a)}{\|\operatorname{Im} f_a\|_{H^1}^2}.$$

Since f_1 is real, it therefore suffices to enclose the five Rayleigh quotients of $f_1, \dots, f_5$.

We enclose the results of these computations as follows. All integrands are circle-invariant polynomials times $|\bar{W}|^{-2}$ or $|\bar{W}|^{-4}$. We replace these factors by 1, which gives an error controlled by $\sup_{S^3} \big||\bar{W}|^2 - 1\big|$, and integrate the polynomials exactly using proposition D.1. The resulting enclosure gives the stated bound. □

3.3 The linear estimate

The linearisation of F at $v = 0$ is given by

$$\begin{aligned} \mathscr{L}h = DF(0)h = \frac{1}{2}\bar{W} \times \Big[& h \times \big(3|\bar{W}|^2 \Delta \bar{W} - \Delta(|\bar{W}|^2 \bar{W})\big) \\ & + \bar{W} \times \big(6(\bar{W} \cdot h)\Delta \bar{W} + 3|\bar{W}|^2 \Delta h - \Delta(2(\bar{W} \cdot h)\bar{W} + |\bar{W}|^2 h)\big)\Big] \\ & + |\bar{W}|^2 \big(\bar{W} \cdot (1 - \Delta)h\big)\bar{W}. \end{aligned} \tag{3.1}$$

3.3.1 An L^2-inverse bound

Fix $l \in \mathbb{Z}_{\geq 0}$ and define the spaces of *low modes* and *high modes* as

$$\mathcal{H}_{\leq l} := \operatorname{span}\{\Psi_k : l_k \leq l\}, \qquad \mathcal{H}_{>l} := \operatorname{span}\{\Psi_k : l_k > l\}.$$

The completion of $\mathcal{H}_{\leq l} \oplus \mathcal{H}_{>l}$ is X, and according to this splitting we write

$$\mathscr{L} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} : \mathcal{H}_{\leq l} \oplus \mathcal{H}_{>l} \to \mathcal{H}_{\leq l} \oplus \mathcal{H}_{>l}.$$

Lower bound for D. Write $\mathscr{R} = \mathscr{L} + \Delta$. The second order term of $\mathscr{R}$ is $(|\bar{W}|^4 - 1)(-\Delta)$. In a local orthonormal frame, write its lower order part as $\sum_i M_i \nabla_i + M_0$, with coefficients obtained by expanding (3.1). Choose $b_1 > 0$ and $b_0 > 0$ such that

$$\sup_{S^3} \sum_i \|M_i\|_{\mathrm{F}}^2 \leq b_1^2, \qquad \sup_{S^3} \|M_0\|_{\mathrm{F}}^2 \leq b_0^2, \tag{3.2}$$

where $\|\cdot\|_{\mathrm{F}}$ is the Frobenius norm. These expressions are independent of the frame. Hence

$$\|\mathscr{R}h\|_{L^2} \leq \eta \|\Delta h\|_{L^2} + b_1 \|\nabla h\|_{L^2} + b_0 \|h\|_{L^2},$$

where η is as in proposition 3.1. The bounds for b_1, b_0 are recorded in table A.1. On $\mathcal{H}_{>l}$, $-\Delta$ has eigenvalues at least $(l+1)(l+3)$. Hence, for $h \in \mathcal{H}_{>l}$:

$$\|h\|_{L^2} \leq \frac{\|\Delta h\|_{L^2}}{(l+1)(l+3)}, \qquad \|\nabla h\|_{L^2}^2 = -\langle h, \Delta h \rangle_{L^2} \leq \frac{\|\Delta h\|_{L^2}^2}{(l+1)(l+3)}.$$

Thus,

$$\|\mathscr{R}h\|_{L^2} \leq \left(\eta + \frac{b_1}{\sqrt{(l+1)(l+3)}} + \frac{b_0}{(l+1)(l+3)}\right) \|\Delta h\|_{L^2}.$$

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Put $\gamma := (1-\eta)(l+1)(l+3) - b_1\sqrt{(l+1)(l+3)} - b_0$ and suppose $\gamma > 0$. Writing Π for the L^2-orthogonal projection onto the high modes, we get

$$\begin{aligned}\|Dh\|_{L^2} &= \|\Pi\mathscr{L}h\|_{L^2}\\ &\geq \|\Delta h\|_{L^2} - \|\Pi\mathscr{R}h\|_{L^2}\\ &\geq \|\Delta h\|_{L^2} - \|\mathscr{R}h\|_{L^2}\\ &\geq \frac{\gamma}{(l+1)(l+3)}\|\Delta h\|_{L^2}\\ &\geq \gamma\|h\|_{L^2},\end{aligned}$$

where in the third-to-last step we used that the orthogonal projection Π has operator norm ≤ 1. On the high modes,

$$D = \big(I + \Pi\mathscr{R}(-\Delta)^{-1}\big)(-\Delta), \qquad \|\Pi\mathscr{R}(-\Delta)^{-1}\|_{L^2\to L^2} \leq 1 - \frac{\gamma}{(l+1)(l+3)} < 1.$$

The Neumann series therefore proves that D is invertible from high-mode H^2 to high-mode L^2, with $\|D^{-1}\|_{L^2\to L^2} \leq \gamma^{-1}$.

Lower bound for A. We compute the entries

$$A_{\alpha\beta} = \int_{S^3} \Psi_\alpha \cdot \mathscr{L}\Psi_\beta \, d\mathrm{vol}, \qquad l_\alpha, l_\beta \leq l,$$

using rigorous quadrature and interval arithmetic and obtain an interval matrix $\mathbf{A} \subset \mathbb{R}^{d\times d}$ containing A, where $d = \dim \mathcal{H}_{\leq l}$.

We compute an upper bound for $\|A^{-1}\|_2$ as follows: write $m(\mathbf{A})$ for its midpoint matrix and compute a non-rigorous candidate for its inverse, $Z \approx m(\mathbf{A})^{-1}$, using floating-point arithmetic. We define $\mathbf{E} := I - Z\mathbf{A}$ and verify

$$\|\mathbf{E}\|_2 \leq \sqrt{\|\mathbf{E}\|_1\|\mathbf{E}\|_\infty} < 1$$

rigorously using interval arithmetic. Here norms of interval matrices denote suprema over their elements. By the Neumann series, every matrix $a \in \mathbf{A}$ is invertible with

$$\|a^{-1}\|_2 \leq \frac{\|Z\|_2}{1 - \|\mathbf{E}\|_2} \leq \frac{\sqrt{\|Z\|_1\|Z\|_\infty}}{1 - \sqrt{\|\mathbf{E}\|_1\|\mathbf{E}\|_\infty}},$$

where the right-hand side is easily computed rigorously. This is the same approach as [35, Lemma 2.1, (2.7)].

Upper bounds for B and C. Let $\mathscr{L}^*$ be the formal adjoint of $\mathscr{L}$. Defining the $d \times d$ matrices

$$G_{\alpha\beta} = \langle \mathscr{L}\Psi_\alpha, \mathscr{L}\Psi_\beta\rangle_{L^2}, \quad G^*_{\alpha\beta} = \langle \mathscr{L}^*\Psi_\alpha, \mathscr{L}^*\Psi_\beta\rangle_{L^2}, \quad l_\alpha, l_\beta \leq l.$$

Orthogonality of low mode and high mode spaces give

$$C^TC = G - A^TA, \qquad BB^T = G^* - AA^T.$$

The matrix A has been computed before, which gives A^TA and AA^T. The matrices G and G^* are readily computed using exact quadrature. The squared norms $\|C\|_2^2$ and $\|B\|_2^2$ are the largest eigenvalues of $G - A^TA$ and $G^* - AA^T$, respectively. We bound them using

$$\lambda_{\max}(S) \leq \min\left\{\|S^J\|_{\mathrm{F}}, \sqrt{\|S^J\|_1\|S^J\|_\infty}\right\}^{1/J}, \quad J \in \mathbb{Z}_{\geq 1}$$

for symmetric positive semidefinite S. In the companion code we choose $J = 32$.

Combining the bounds using the Schur complement. All operator norms below are induced by L^2. Using the same approximate inverse Z, we also bound

$$\|A^{-1}B\| \leq \frac{\|ZB\|_2}{1 - \|\mathbf{E}\|_2} \quad \text{using} \quad \|ZB\|_2^2 = \|ZBB^TZ^T\|_2 = \|Z(G^* - AA^T)Z^T\|_2,$$

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where the finite-dimensional matrices on the right hand side have been computed before. Let $S := A - BD^{-1}C : \mathcal{H}_{\le l} \to \mathcal{H}_{\le l}$ be the Schur complement of D. For $u \in \mathcal{H}_{\le l}$,

$$\begin{aligned}\|Su\|_{L^2} &\ge \frac{\|A^{-1}Su\|_{L^2}}{\|A^{-1}\|} \\ &= \frac{\|u - A^{-1}BD^{-1}Cu\|_{L^2}}{\|A^{-1}\|} \\ &\ge \frac{1 - \|A^{-1}B\|\|D^{-1}\|\|C\|}{\|A^{-1}\|}\|u\|_{L^2}.\end{aligned}$$

Suppose the numerator is positive. Since $\mathcal{H}_{\le l}$ is finite-dimensional, S is invertible. The block inverse formula gives

$$\begin{aligned}\mathscr{L}^{-1} &= \begin{pmatrix} S^{-1} & -S^{-1}BD^{-1} \\ -D^{-1}CS^{-1} & D^{-1} + D^{-1}CS^{-1}BD^{-1} \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & D^{-1} \end{pmatrix} + \begin{pmatrix} I \\ -D^{-1}C \end{pmatrix} S^{-1} \begin{pmatrix} I & -BD^{-1} \end{pmatrix}.\end{aligned}$$

Taking norms, using the L^2-orthogonality of the splitting and the preceding estimate, gives

$$\|\mathscr{L}^{-1}\|_{L^2 \to L^2} \le \|D^{-1}\| + \frac{\|A^{-1}\|\sqrt{(1 + \|B\|^2\|D^{-1}\|^2)\,(1 + \|C\|^2\|D^{-1}\|^2)}}{1 - \|A^{-1}B\|\|D^{-1}\|\|C\|}.$$

3.3.2 A priori estimate

For $h \in X$, recalling that $\eta < 1$, we have

$$\begin{aligned}\|\Delta h\|_{L^2} &\le \|\mathscr{L}h\|_{L^2} + \|\mathscr{R}h\|_{L^2} \\ &\le \frac{1}{1-\eta}\left(\|\mathscr{L}h\|_{L^2} + b_1\|\nabla h\|_{L^2} + b_0\|h\|_{L^2}\right) \\ &\le \frac{1}{1-\eta}\left(\|\mathscr{L}h\|_{L^2} + b_1\sqrt{\|h\|_{L^2}\|\Delta h\|_{L^2}} + b_0\|h\|_{L^2}\right) \\ &\le \frac{\|\mathscr{L}h\|_{L^2}}{1-\eta} + \frac{1}{2}\|\Delta h\|_{L^2} + \left(\frac{b_0}{1-\eta} + \frac{b_1^2}{2(1-\eta)^2}\right)\|h\|_{L^2}.\end{aligned}$$

Here the first step uses the definition of $\mathscr{R}$; the second uses (3.2), the preceding inequality, and rearrangement; the third uses integration by parts and Cauchy–Schwarz; and the fourth uses Young's inequality.

Thus, absorbing $\frac{1}{2}\|\Delta h\|_{L^2}$, we obtain

$$\|h\|_X \le \|\Delta h\|_{L^2} + \|h\|_{L^2} \le \frac{2}{1-\eta}\|\mathscr{L}h\|_{L^2} + \left(1 + \frac{2b_0}{1-\eta} + \frac{b_1^2}{(1-\eta)^2}\right)\|h\|_{L^2},$$

where the first inequality follows from $\|h\|_X = \|(1-\Delta)h\|_{L^2}$ and the triangle inequality.

3.3.3 Injectivity estimate

Combining the estimates from the preceding two subsubsections, we obtain, for $h \in X$,

$$\begin{aligned}\|h\|_X &\le \frac{2}{1-\eta}\|\mathscr{L}h\|_{L^2} + \left(1 + \frac{2b_0}{1-\eta} + \frac{b_1^2}{(1-\eta)^2}\right)\|h\|_{L^2} \\ &\le \left[\frac{2}{1-\eta} + \left(1 + \frac{2b_0}{1-\eta} + \frac{b_1^2}{(1-\eta)^2}\right)\|\mathscr{L}^{-1}\|_{L^2 \to L^2}\right]\|\mathscr{L}h\|_Y.\end{aligned}$$

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3.4 Higher order estimate

Write ∇_i for differentiation along a local orthonormal frame; all contractions below are frame-independent. The higher order terms of (2.5) are

$$\begin{aligned}\mathscr{N} : X &\longrightarrow Y,\\ v &\longmapsto F(v) - F(0) - \mathscr{L}v\\ &\quad = 6\mathcal{T}(\bar{W},\bar{W},v,v) + 4\mathcal{T}(\bar{W},v,v,v) + \mathcal{T}(v,v,v,v),\\ \mathcal{T}(a_1,a_2,a_3,a_4) &= \frac{1}{24}\bar{W}\times\sum_{\sigma\in S_4}\Big[(a_{\sigma(1)}\cdot a_{\sigma(2)})(a_{\sigma(3)}\times\Delta a_{\sigma(4)}) - 2\sum_i (a_{\sigma(1)}\cdot\nabla_i a_{\sigma(2)})(a_{\sigma(3)}\times\nabla_i a_{\sigma(4)})\Big],\end{aligned}$$

for $a = (a_1,a_2,a_3,a_4) \in X^{\oplus 4}$.

Proposition 3.3. *For $v, w \in X$ with $\|v\|_X, \|w\|_X \le r$, we have*

$$\|\mathscr{N}(v) - \mathscr{N}(w)\|_Y \le C_8 r(3\|\bar{W}\|_X^2 + 3r\|\bar{W}\|_X + r^2)\|v-w\|_X,$$

where

$$C_8 = 4\|\bar{W}\|_\infty(c_\infty^3 + 2c_\infty^2 c_4^2).$$

Proof. We have

$$\big\|(a_{\sigma(1)}\cdot a_{\sigma(2)})(a_{\sigma(3)}\times\Delta a_{\sigma(4)})\big\|_{L^2} \le \|a_{\sigma(1)}\cdot a_{\sigma(2)}\|_{L^\infty}\|a_{\sigma(3)}\|_{L^\infty}\|\Delta a_{\sigma(4)}\|_{L^2} \le c_\infty^3\prod_{j=1}^4\|a_j\|_X,$$

where in the first step we pulled out the maxima; in the second we used Cauchy–Schwarz and proposition 2.4. Analogously one finds that the sum over i in the definition of $\mathcal{T}$ is bounded in L^2 by $c_\infty^2 c_4^2\prod_{j=1}^4\|a_j\|_X$. Hence,

$$\|\mathcal{T}(a_1,a_2,a_3,a_4)\|_Y \le \frac{C_8}{4}\prod_{j=1}^4\|a_j\|_X.$$

Telescoping as

$$\mathcal{T}(v,v,v,v) - \mathcal{T}(w,w,w,w) = \mathcal{T}(v-w,v,v,v) + \mathcal{T}(w,v-w,v,v) + \mathcal{T}(w,w,v-w,v) + \mathcal{T}(w,w,w,v-w),$$

applying the preceding bound, and the analogous two- and three-term telescopes to the other terms of $\mathscr{N}$, give

$$\|\mathscr{N}(v) - \mathscr{N}(w)\|_Y \le \frac{C_8}{4}(12r\|\bar{W}\|_X^2 + 12r^2\|\bar{W}\|_X + 4r^3)\|v-w\|_X.$$

Factoring out $4r$ proves the claim. □

3.5 Energy estimate

Proposition 3.4. *Let $W, \bar{W} \in X$ with $\|W-\bar{W}\|_X \le r$, and suppose that $u = W/|W|$ is harmonic. Assume*

$$\min_{S^3}|\bar{W}| - c_\infty r \ge C_1 > c_\infty r.$$

Then

$$E\left(\frac{\bar{W}}{|\bar{W}|}\right) - \frac{c_5^2 r^2}{2C_1^2}\left(1 + \frac{4c_\infty\|\bar{W}\|_X}{C_1}\right)^2 \le E(u),$$

$$E(u) \le \frac{E\left(\frac{\bar{W}}{|\bar{W}|}\right)(C_1 + c_\infty r)^2}{C_1^2 - 3c_\infty^2 r^2 + 2c_\infty r C_1}.$$

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Proof. We now bound the energy of u by that of $\bar{u} = \bar{W}/|\bar{W}|$. Write $v = W - \bar{W}$ and $w = \bar{u} - u$. Then, since u is harmonic,

$$E(\bar{u}) = E(u) + \tfrac{1}{2}\|dw\|_{L^2}^2 - \tfrac{1}{2}\int |du|^2|w|^2.$$

Suppose that $||w||_{C^0}^2 < 1$. Rearranging the above slightly, we can obtain two inequalities

$$E(\bar{u}) - \tfrac{1}{2}\|dw\|_{L^2}^2 \;\le\; E(u) \;\le\; \frac{E(\bar{u})}{1 - \|w\|_{C^0}^2}.$$

To prove the proposition, there are now two steps:

1. Obtain an upper bound on $\|w\|_{C^0}^2$. To do this, we use the elementary inequality $|a/|a| - b/|b|| \le 2|a-b|/|a|$, to get that
$$\|w\|_{C^0} \le \frac{2c_\infty r}{\min_{S^3}|\bar{W}|} \le \frac{2c_\infty r}{C_1 + c_\infty r} < 1.$$
This validates the assumption that $\|w\|_{C^0}^2 < 1$.

2. The lower bound. Let $\varphi(X) = X/|X|$ denote the radial map on $X \in \mathbb{R}^3$. In the following, we will use the following elementary estimates for φ:
$$|d\varphi_X| \le \frac{1}{|X|} \qquad\qquad |d^2\varphi_X| \le \frac{4}{|X|^2} \tag{3.3}$$
Now, to obtain a pointwise bound on dw, we write
$$dw = d\varphi_{\bar{W}}(d\bar{W}) - d\varphi_W(d\bar{W} + dv) = -\,d\varphi_W(dv) + \big[d\varphi_{\bar{W}} - d\varphi_W\big](d\bar{W}).$$
To bound the first term, then by (3.3) and the inequality $C_1 \le |\bar{W}| - |v| \le |W|$, we have
$$|d\varphi_W(dv)| \le \frac{|dv|}{|W|} \le \frac{|dv|}{C_1}. \tag{3.4}$$
For the second term, by the fundamental theorem of calculus, we have that
$$d\varphi_W - d\varphi_{\bar{W}} = \int_0^1 d^2\varphi_{\bar{W}+tv}(v,\cdot)\,dt,$$
Then, as $C_1 \le |\bar{W}| - |v| \le |\bar{W} + tv|$, for all $t \in [0,1]$, using the second estimate in (3.3), we have
$$|\big[d\varphi_{\bar{W}} - d\varphi_W\big](d\bar{W})| \le \frac{4\,|v|\,|d\bar{W}|}{C_1^2}. \tag{3.5}$$
Putting these both together, and integrating over S^3, we get
$$\|dw\|_{L^2} \le \frac{\|dv\|_{L^2}}{C_1} + \frac{4\|v\|_{C^0}\|d\bar{W}\|_{L^2}}{C_1^2} \le \frac{c_5 r}{C_1}\left(1 + \frac{4\,c_\infty\,\|\bar{W}\|_X}{C_1}\right),$$
where we have used Proposition 2.5 in the second inequality.

□

4 Conclusion of the proof of the main theorems

Proof of theorem 1.1. Existence and Hopf invariant. The estimates in §3.3 and propositions 3.1 and 3.3 verify the hypotheses of theorem 2.19 with M , $\varepsilon = C_2$, r from table A.1 and
$$k = C_8 r(3\|\bar{W}\|_X^2 + 3r\|\bar{W}\|_X + r^2).$$

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Hence, there exists $v \in X$ with $\|v\|_X \leq r$ and $F(v) = 0$.

Set $W = \bar{W} + v$ and let $V = V_q$ be the reference map from (2.8). Since $|V| \leq 1$ by lemma 2.12, we have

$$W \cdot V \geq \bar{W} \cdot V - |v||V| \geq C_5 - c_\infty r > 0, \tag{4.1}$$

and $\min |\bar{W}| \geq C_5 > c_\infty r$. Thus proposition 2.6 gives a smooth G-equivariant harmonic map $u = W/|W|$. Since $u \cdot V > 0$, proposition 2.15 gives $H(u) = H(V/|V|) = q$.

Energy. The certified bound in table A.1 gives $C_1 > c_\infty r$. Applying proposition 3.4 and the approximate energy bounds in proposition 3.1 gives $C_9 \leq E(u) \leq C_{10}$, where

$$C_9 := C_3 - \frac{c_5^2 r^2}{2C_1^2}\left(1 + \frac{4c_\infty \|\bar{W}\|_X}{C_1}\right)^2,$$
$$C_{10} := \frac{C_4(C_1 + c_\infty r)^2}{C_1^2 - 3c_\infty^2 r^2 + 2c_\infty r C_1}.$$

The resulting numerical bounds are given in table A.1. These bounds exclude $\frac{E(u)}{8\pi^2}$ from being an integer, so u cannot have the form $u = R \circ \pi \circ A$.

Index. Given the nine test fields $\xi_a \in \mathbb{H}^1(S^3, \mathbb{R}^3)$ from proposition 3.2, we let δ be as in proposition 2.17. Evaluating the formula for δ in interval arithmetic, using r and C_1 from table A.1, the bound $\|\bar{W}\|_X \leq C_6$ from proposition 3.1, and the exact constants c_∞, c_4, β of proposition 2.4, gives $\delta \leq 2.798 \cdot 10^{-4}$, see also table A.1 so $\lambda^* + \delta \leq -0.04109 + 0.00028 < 0$. Since the fields are orthogonal for Q_W, and the H^1-inner product, the inequality $\lambda^* + \delta < 0$ makes $G + \delta H$ negative definite. Thus lemma 2.18 and proposition 2.16 give $\operatorname{Ind} Q_u \geq 9$.

□

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Data availability

Self-contained code for the numerics used in this proof, as well as certificates for proof steps, are available at this paper's respective GitHub repository: `github.com/edhirst/eells_counterexample`

AI usage

This work started with an application of physics-informed neural networks to model the map between S^3 and S^2; aligning with ideas introduced in [24]. In this construction, to ensure the Hopf invariant was fixed throughout the learning process, the neural architecture modelled a homotopy map from the standard Hopf-related construction in each class at $t = 0$ to an open endpoint at $t = 1$. This AI architecture was then trained with the goal of minimising the harmonic energy at the $t = 1$ endpoint to identify potentially new harmonic maps. Training produced many promising candidates across Hopf invariant classes, and studying these approximations lead to the identification of the symmetry used in the numerical approximant function class used in this proof.

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With this information, Claude (Anthropic) then assisted with building the full candidate function, writing code for the numerics, and drafting key stages of this computer-assisted proof. Codex (OpenAI) was later used to assess and review what was produced. The mathematical paper was written entirely by the authors, who retain full responsibility for the mathematical content and computational results.

A Numerical constants

In this appendix, we list the numerical constants needed for the proof of the main theorems in §4. We include the choice of cut-off l of §3.3. As required by §4 for theorem 1.1, in every column $Mk < 1$, $M(\varepsilon + kr) \le r$, $C_5 - c_\infty r > 0$ and $\|w\|_{C^0} < 1$, moreover, for the index computation $\lambda^* + \delta < 0$, see proposition 2.17. The constants $c_\infty, c_4, c_5, \beta$ are the exact values coming from our Sobolev norms in propositions 2.4 and 2.5; the certificates use interval enclosures of them, which lie in the intervals shown.

Hopf invariant q	2	3	4	5	6
$\deg \bar{W}$	40	80	112	152	184
l	40	40	60	60	80
C_2	$\le 1.987 \cdot 10^{-8}$	$\le 5.179 \cdot 10^{-11}$	$\le 3.855 \cdot 10^{-11}$	$\le 4.657 \cdot 10^{-11}$	$\le 6.482 \cdot 10^{-11}$
C_3	≥ 125.2744885849	≥ 155.6036384307	≥ 177.7689807885	≥ 195.1505049029	≥ 209.4236634609
C_4	≤ 125.2744885857	≤ 155.6036384337	≤ 177.7689807937	≤ 195.1505049116	≤ 209.4236634726
C_5	≥ 0.2499	≥ 0.2450	≥ 0.1917	≥ 0.1347	≥ 0.08728
C_8	≤ 0.03534	≤ 0.03534	≤ 0.03534	≤ 0.03534	≤ 0.03534
C_9	≥ 125.2744885848	≥ 155.6036384307	≥ 177.7689807885	≥ 195.1505049029	≥ 209.4236634609
C_{10}	≤ 125.2744885858	≤ 155.6036384337	≤ 177.7689807937	≤ 195.1505049116	≤ 209.4236634726
C_1	0.99999986	0.99999999	0.99999999	0.99999999	0.99999998
C_6	69.058	99.765	130.98	162.42	193.99
C_7	1.00001	1.00001	1.00001	1.00001	1.00001
η	$2.338 \cdot 10^{-9}$	$1.161 \cdot 10^{-8}$	$4.118 \cdot 10^{-8}$	$1.207 \cdot 10^{-7}$	$2.097 \cdot 10^{-7}$
b_1	≤ 11.41	≤ 17.17	≤ 22.93	≤ 28.69	≤ 34.46
b_0	≤ 5.751	≤ 9.562	≤ 16.45	≤ 25.50	≤ 36.64
γ	≥ 1278	≥ 1032	≥ 2405	≥ 2039	≥ 3861
$\|A^{-1}\|$	≤ 0.1476	≤ 0.1363	≤ 0.1276	≤ 0.1225	≤ 0.1192
$\|B\|$	≤ 160.7	≤ 261.0	≤ 490.3	≤ 648.1	≤ 994.1
$\|C\|$	≤ 138.6	≤ 181.3	≤ 373.8	≤ 433.8	≤ 715.5
$\|A^{-1}B\|$	≤ 0.1038	≤ 0.1754	≤ 0.1511	≤ 0.2067	≤ 0.1793
$\|\mathscr{L}^{-1}\|_{L^2\to L^2}$	≤ 0.1521	≤ 0.1482	≤ 0.1353	≤ 0.1380	≤ 0.1298
$\|\mathscr{L}^{-1}\|_{Y\to X}$	$\le M = 23.67$	$\le M = 48.63$	$\le M = 77.67$	$\le M = 122.7$	$\le M = 165.6$
r	$9.5 \cdot 10^{-7}$	$5.1 \cdot 10^{-9}$	$6 \cdot 10^{-9}$	$1.2 \cdot 10^{-8}$	$2.2 \cdot 10^{-8}$
Mk	≤ 0.0114	$\le 2.62 \cdot 10^{-4}$	$\le 8.48 \cdot 10^{-4}$	≤ 0.00412	≤ 0.0146
$M(\varepsilon + kr)$	$\le 4.82 \cdot 10^{-7}$	$\le 2.52 \cdot 10^{-9}$	$\le 3 \cdot 10^{-9}$	$\le 5.76 \cdot 10^{-9}$	$\le 1.11 \cdot 10^{-8}$
$\|w\|_{C^0}$	$\le 2.69 \cdot 10^{-7}$	$\le 1.45 \cdot 10^{-9}$	$\le 1.7 \cdot 10^{-9}$	$\le 3.4 \cdot 10^{-9}$	$\le 6.23 \cdot 10^{-9}$
δ	$\le 2.798 \cdot 10^{-4}$	–	–	–	–
λ^*	≤ -0.04109	–	–	–	–
c_∞	$\sqrt{\frac{\pi^2/6 - 5/4}{2\pi^2}} \in [0.1414482009672520, 0.1414482009672527]$				
c_4	$\left(\frac{2}{9\pi^2}\right)^{1/4} \in [0.3873663890421022, 0.3873663890421032]$				
c_5	$\frac{2\sqrt{2}}{9} \in [0.3142696805273541, 0.3142696805273548]$				
β	$(2\pi^2)^{-1/4} \in [0.4744249983287935, 0.4744249983287952]$				

Table A.1: Constants of the certificate for the approximate solutions $\bar{W} = \bar{W}_q$ of Hopf invariant $q = 2, \dots, 6$ in theorem 1.1. Upper bounds are rounded up and lower bounds down.

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B The approximate solutions

The approximate solutions $\bar{W} = \bar{W}_q$ are given by equivariant polynomial maps $S^3 \to \mathbb{R}^3$ in the span of the basis of proposition 2.3, of the degree given in table A.1. Its leading coefficients are listed in table B.1; the coefficients vary smoothly with q.

In the table below, we distinguish the two types of fields in the basis of proposition 2.3, which we will call type U, V:

$$(\operatorname{Re} Y_{q(n-2),n,j}, \operatorname{Im} Y_{q(n-2),n,j}, 0) \qquad \sqrt{2}(0, 0, \operatorname{Re} Y_{qn,n,j})$$

with n even, and $n \geq 1$ odd, respectively.

For every q, more than 99.99% of the L^2-norm of the approximate solution $\|\bar{W}_q\|^2_{L^2}$ comes from the fields of type U with $n = 0, 2$ and of type V with $n = 1$.

type	n	j	$q=2$	$q=3$	$q=4$	$q=5$	$q=6$
U	0	0	−1.525663	−1.023630	−0.744217	−0.571476	−0.456312
U	0	1	0.291036	0.343169	0.327313	0.295575	0.262980
U	0	2	−0.063119	−0.122853	−0.149767	−0.156980	−0.154470
U	0	3	0.015490	0.046593	0.070918	0.085308	0.092249
U	0	4	−0.004064	−0.018564	−0.034639	−0.047342	−0.055932
U	0	5	0.001097	0.007679	0.017388	0.026785	0.034396
U	0	6	−0.000301	−0.003259	−0.008930	−0.015421	−0.021433
U	0	7	0.000083	0.001407	0.004670	0.009015	0.013518
U	2	0	3.478069	3.719205	3.794535	3.822473	3.834396
U	2	1	−0.898244	−1.325303	−1.500491	−1.576199	−1.611709
U	2	2	0.205195	0.529395	0.730052	0.837857	0.895588
U	2	3	−0.049602	−0.209652	−0.367253	−0.476414	−0.544815
U	2	4	0.012938	0.083949	0.185306	0.275383	0.341920
U	2	5	−0.003515	−0.034444	−0.093954	−0.159794	−0.216736
U	2	6	0.000971	0.014507	0.048120	0.093011	0.137848
U	2	7	−0.000271	−0.006241	−0.024974	−0.054421	−0.087877
U	2	8	0.000076	0.002725	0.013140	0.032072	0.056196
U	2	9	−0.000021	−0.001201	−0.007000	−0.019059	−0.036091
U	2	10	0.000006	0.000533	0.003766	0.011423	0.023300
U	2	11	−0.000002	−0.000237	−0.002042	−0.006901	−0.015129
U	4	0	−0.012727	−0.008865	−0.005399	−0.003338	−0.002147
V	1	0	2.021640	1.464655	1.081160	0.823037	0.644811
V	1	1	−0.519577	−0.637786	−0.612132	−0.547986	−0.479881
V	1	2	0.124740	0.261339	0.321739	0.334523	0.324242
V	1	3	−0.032001	−0.106537	−0.166037	−0.199211	−0.212729
V	1	4	0.008680	0.044216	0.085636	0.117712	0.137966
V	1	5	−0.002410	−0.018788	−0.044528	−0.069560	−0.089123
V	1	6	0.000675	0.008145	0.023429	0.041290	0.057582
V	1	7	−0.000190	−0.003580	−0.012478	−0.024677	−0.037306
V	1	8	0.000054	0.001587	0.006717	0.014861	0.024273
V	1	9	−0.000015	−0.000708	−0.003646	−0.009017	−0.015874
V	1	10	0.000004	0.000317	0.001992	0.005507	0.010436
V	3	0	−0.011423	−0.007876	−0.004977	−0.003238	−0.002198
$\deg \bar{W}_q$			40	80	112	152	184
number of coefficients			217	630	969	1482	1847

Table B.1: Leading coefficients a_k of the approximate solutions $\bar{W}_q = \sum_k a_k \Psi_k$ in the orthonormal basis of Proposition 2.3. Listed are all basis fields whose coefficient has modulus at least 0.01 for some q, rounded to 6 decimals. The complete lists, with the exact IEEE double coefficients used by the certificate, are the files `data/Wbar_qq_DD.json` of the repository.

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C The test fields for the index

In this appendix, we give a list of negative directions ξ_a for our approximate solution $\bar{W}$ with Hopf invariant 2, which were found using computer-assisted methods. To understand these negative directions somewhat systematically, we use the action of the subgroup $G' := U(1) \times \mathbb{Z}_2 \subset G$ generated by $e^{i\theta}$ and $e^{i\pi}\varepsilon$ on $H^1(S^3, \mathbb{R}^3)$. Its action on fields is $(g \cdot \xi)(x) = g\,\xi(g^{-1}x)$. By the Peter-Weyl theorem, this induces an orthogonal splitting of $H^1(S^3, \mathbb{R}^3)$ into irreducible representations.

Since $\bar{W}$ is G'-equivariant, $Q_{\bar{W}}$ and the H^1 inner product are G'-invariant. Distinct isotypic components are therefore orthogonal by Schur's lemma.

We can write a basis of isotypic G'-representations explicitly over $\mathbb{C}$. For $Y = Y_{m,n,j}$ put

$$P_{m,n,j} = \tfrac{1}{\sqrt{2}}(Y, -iY, 0), \qquad M_{m,n,j} = \tfrac{1}{\sqrt{2}}(Y, iY, 0), \qquad T_{m,n,j} = (0, 0, Y). \tag{C.1}$$

By Lemma 2.2 the action of G' on this basis is given by $e^{i\theta}.\xi = e^{i\nu\theta}\xi$ and $(e^{i\pi}\varepsilon) \cdot \xi = p\,\xi$, where $\nu = 2n - m - 4,\ 2n - m + 4,\ 2n - m$ and $p = (-1)^l,\ (-1)^l,\ -(-1)^l$ respectively.

Let $S_{\nu,p}$ be the complex linear span of P, M, T in (C.1) with weights (ν, p) for the G'-action. Complex conjugation maps $S_{\nu,p}$ to $S_{-\nu,p}$. Table C.1 lists the five fields $f_1, \dots, f_5$ for $q = 2$, which give the nine negative directions $\xi_1 = f_1$, $\operatorname{Re} f_a$, $\operatorname{Im} f_a$ $(a = 2, \dots, 5)$.

For $f \in S_{\nu,p}$ with $\nu \neq 0$, rotation by $\pi/(2\nu)$ sends $(\operatorname{Re} f, \operatorname{Im} f)$ to $(-\operatorname{Im} f, \operatorname{Re} f)$. Hence, $\operatorname{Re} f$ and $\operatorname{Im} f$ are orthogonal and have equal quadratic values for both forms. Hence, the nine given fields are nonzero and pairwise orthogonal.

$f_1 \in S_{0,-}$		$f_2 \in S_{1,-}$		$f_3 \in S_{2,-}$		$f_4 \in S_{2,+}$		$f_5 \in S_{3,-}$	
$T_{0,0,0}$	−100	$T_{1,1,0}$	−89	$M_{0,-1,0}$	100	$T_{0,1,0}$	100	$M_{1,0,0}$	100
$T_{0,0,1}$	−56	$M_{-1,-2,0}$	17	$T_{-2,0,0}$	41	$M_{2,0,0}$	−66	$T_{-1,1,0}$	−76
$M_{2,-1,0}$	39	$M_{3,0,0}$	100	$M_{0,-1,1}$	−48	$T_{0,1,1}$	1	$M_{1,0,1}$	−15
$P_{-2,1,0}$	39	$P_{-1,2,0}$	16	$P_{0,3,0}$	−64	$M_{-2,-2,0}$	−22	$T_{-1,1,1}$	−11
$T_{0,0,2}$	12	$T_{-3,-1,0}$	46	$T_{-2,0,1}$	−35	$M_{2,0,1}$	−2	$M_{-3,-2,0}$	23
$M_{-2,-3,0}$	10	$T_{1,1,1}$	5	$T_{2,2,0}$	−42	$P_{-2,2,0}$	−23	$M_{1,0,2}$	−9
$M_{2,-1,1}$	−2	$M_{-1,-2,1}$	−14	$M_{0,-1,2}$	19	$T_{-4,-1,0}$	−25	$P_{-3,2,0}$	33
$P_{-2,1,1}$	−2	$M_{3,0,1}$	−8	$M_{4,1,0}$	27	$T_{0,1,2}$	−8	$P_{1,4,0}$	−18
$P_{2,3,0}$	10	$P_{-5,0,0}$	−63	$P_{-4,1,0}$	29	$M_{-2,-2,1}$	8	$T_{-5,-1,0}$	30
$T_{-4,-2,0}$	9	$P_{-1,2,1}$	−14	$P_{0,3,1}$	27	$M_{2,0,2}$	3	$T_{-1,1,2}$	12
$T_{0,0,3}$	1	$T_{-3,-1,1}$	−17	$T_{-2,0,2}$	14	$P_{-6,0,0}$	26	$T_{3,3,0}$	−8
$T_{4,2,0}$	9	$T_{1,1,2}$	3	$T_{2,2,1}$	16	$P_{-2,2,1}$	8	$M_{-3,-2,1}$	−8
$M_{-2,-3,1}$	−4	$M_{-1,-2,2}$	5	$M_{0,-1,3}$	−6	$P_{2,4,0}$	−1	$M_{1,0,3}$	5
$M_{2,-1,2}$	−1	$P_{-5,0,1}$	14	$M_{4,1,1}$	−8	$T_{-4,-1,1}$	7	$M_{5,2,0}$	3
$M_{6,1,0}$	−6	$P_{-1,2,2}$	5	$P_{-4,1,1}$	−8	$T_{0,1,3}$	3	$P_{-7,0,0}$	−31
$P_{-6,-1,0}$	−6	$P_{3,4,0}$	1	$P_{0,3,2}$	−8	$T_{4,3,0}$	−1	$P_{-3,2,1}$	−11
$P_{-2,1,2}$	−1	$T_{-3,-1,2}$	4	$T_{-2,0,3}$	−4	$M_{-2,-2,2}$	−2	$P_{1,4,1}$	6
$P_{2,3,1}$	−4	$T_{1,1,3}$	−1	$T_{2,2,2}$	−4	$M_{2,0,3}$	−1	$T_{-5,-1,1}$	−7
$T_{-4,-2,1}$	−2	$T_{5,3,0}$	1			$M_{6,2,0}$	1	$T_{-1,1,3}$	−4
$T_{0,0,4}$	−1					$P_{-6,0,1}$	−5	$T_{3,3,1}$	2
$T_{4,2,1}$	−2					$P_{-2,2,2}$	−2		
≤ -0.4693		≤ -0.04850		≤ -0.05998		≤ -0.2957		≤ -0.04109	

Table C.1: The test fields of proposition 3.2 ($q = 2$): $f_a = \sum c\,\Phi$ with Gaussian-integer coefficients c and modes $\Phi \in \{P_{m,n,j}, M_{m,n,j}, T_{m,n,j}\}$ of (C.1) in the sector $S_{\nu,p}$ of f_a. They are rounded eigenvectors of the Galerkin matrix of $Q_{\bar{W}}$ on the modes of degree ≤ 8 of each sector. The last row is the certified bound on $Q_{\bar{W}}(\xi)/\|\xi\|^2_{H^1}$ for $\xi = f_1$, respectively $\xi = \operatorname{Re} f_a, \operatorname{Im} f_a$ $(a \geq 2)$; its maximum is λ^*.

D Exact quadrature

Throughout the paper, we computed many integrals of spherical harmonics. Numerically, these calculations are implemented as *exact quadrature*. That is: if the number of quadrature points is large compared to the polynomial degree, then quadrature gives the *exact* value of the integral. See [43, Sections 1 and 3] for an overview over this standard approach.

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In our case, we are integrating functions $S^3 \to \mathbb{C}$ that are invariant under a circle action. This reduces the problem to two-dimensional quadrature. We record here our exact quadrature rule.

Fix integers $N, M \geq 2$. For $0 \leq i \leq N$ and $0 \leq j < M$, set

$$s_i = \frac{1 - \cos(\pi i/N)}{2}, \qquad x_{ij} = \left(\sqrt{s_i}, \sqrt{1 - s_i}\, e^{2\pi \mathrm{i} j/M}\right) \in S^3.$$

Let $\ell_i \in \mathbb{R}[s]$ be the (Lagrange interpolation) polynomial of degree at most N satisfying $\ell_i(s_k) = \delta_{ik}$ for $0 \leq k \leq N$. To each x_{ij} we assign the weight

$$\omega_i = \frac{2\pi^2}{M} \int_0^1 \ell_i(s)\, ds,$$

independent of j. This corresponds to Clenshaw–Curtis quadrature in s [43, p. 135] and the trapezoidal rule in the angle variable [44, p. 394, equation (3.2)]. The exact quadrature result is then:

Proposition D.1. *Let $q \geq 2$ be an integer and let $f : S^3 \to \mathbb{C}$ be invariant under $(z, w) \mapsto (e^{\mathrm{i}\theta} z, e^{-q\mathrm{i}\theta} w)$. Suppose that f admits a polynomial extension to $\mathbb{R}^4$ of degree at most p. If $p \leq 2N$ and $p < (q+1)M$, then*

$$\int_{S^3} f \, d\mathrm{vol} = \sum_{i=0}^{N} \sum_{j=0}^{M-1} \omega_i f(x_{ij}).$$

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